2. Characterization of low molecular mass compounds

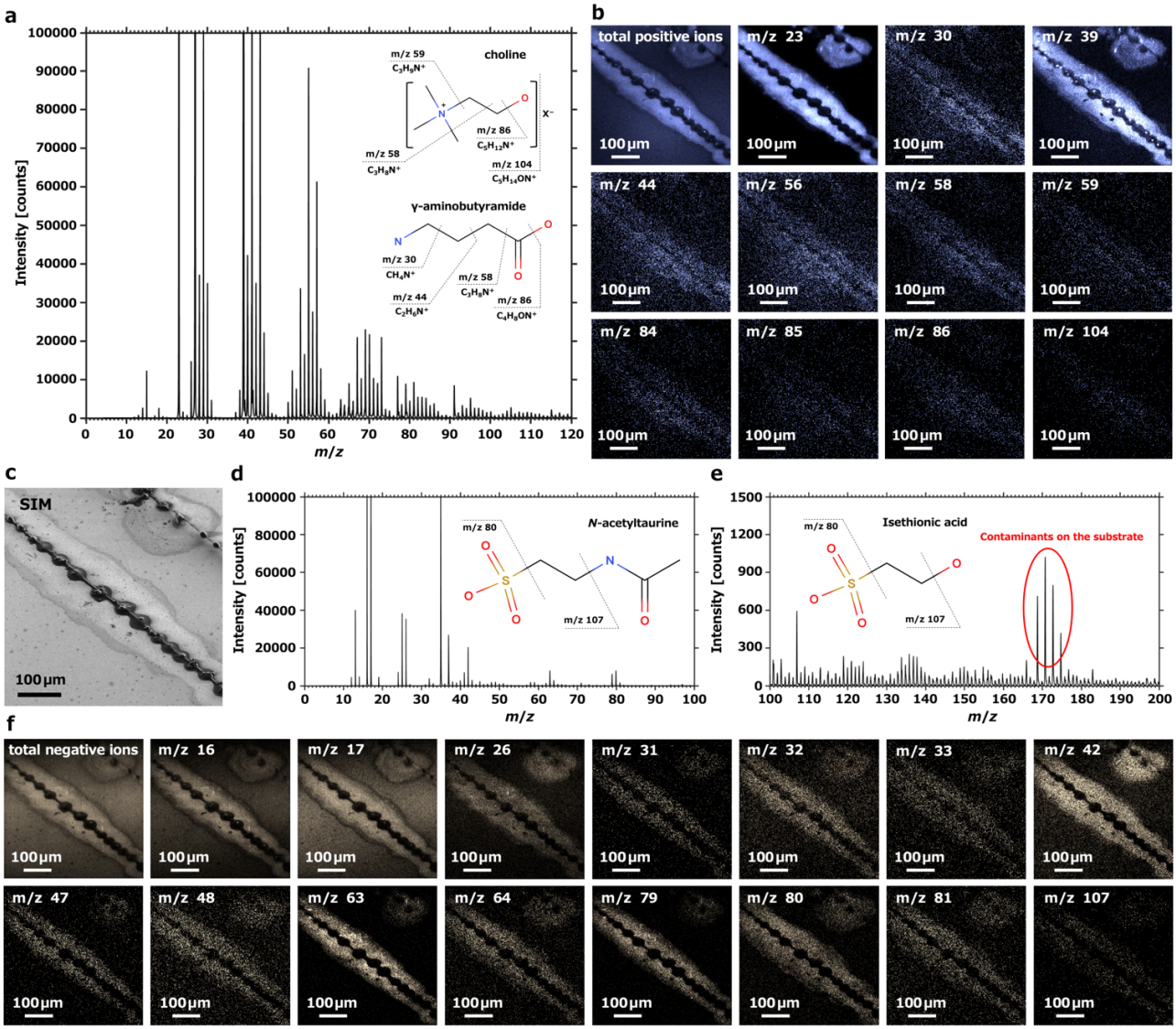

Figure 2 | Observations of aggregate glue droplets using TOF-SIMS. a, Positive ions mass spectrum of *m/z* 1 to 120, and **b,** secondary ion mapped images of positive ions. **c,** Scanning ion microscopy (SIM) image of the aggregate glue droplets attached to the substrate. Negative ions mass spectra of **d,** *m/z* 1 to 100 and **e,** *m/z* 100 to 200. **f,** Secondary ion mapped images of negative ions. The mapped color is a pseudo-color.

TOF-SIMS is an excellent surface analysis method. In particular, TOF-SIMS with a focused Ga^+ ion beam is effective for analyzing small particles [22]. Hence, here the sample of attached aggregate glue

droplets was mapped its element distribution using a home-made TOF-SIMS. We measured positive and negative ions separately with a mass spectrometry range of *m*/*z* 1 to 300 (see Supporting Information Figs. S1 and S2). First, we observed the scanning ion microscopy image of the sample (see Fig. 2c). In the scanning ion microscopy image, two capture spiral lines were observed, the diagonal one in the field of view was completely attached to the substrate, and the upper right one was partially suspended. The attached aggregate glue droplets took on a "fried egg" structure. From the contrast of scanning ion microscopy image, the "fried egg" structure consists of a darker central area and a spread brighter area.

In all mapping images, we selected images which intensity distribution can coincide with the "fried egg" structure and showed them in Figs. 3d and 3f. All mapping images of positive and negative ions show in Supporting Information Figs. S3 and S4.

In the positive secondary ion mass spectrum and mapped images in Figs. 3a and 3b, *m*/*z* 23 and *m*/*z* 39 are assigned to Na^+ and K^+, respectively. This is consistent with our previous analysis of the inorganic matter of the aggregate glue droplets [20,23]. Other fragment ions are attributed as follows: *m*/*z* 30 corresponds to $(H_2N–CH_2)^+$, *m*/*z* 44 to $(H_2N–CH_2–CH_2)^+$, *m*/*z* 58 to $(H_2N–CH_2–CH_2–CH_2)^+$ and/or $((CH_3)_2N=CH_2)^+$, *m*/*z* 59 to $((CH_3)_3N)^+$, *m*/*z* 84 to $(C_4H_6NO)^+$, *m*/*z* 86 to $(H_2N–CH_2–CH_2–CH_2–C=O)^+$, $((CH_3)_3N–CH=CH_2)^+$ and/or $(C_4H_8NO)^+$, and *m*/*z* 104 to $((CH_3)_3N–CH_2–CH_2–OH)^+$.

The associated fragment ions at *m*/*z* 30 $(H_2N–CH_2)^+$, *m*/*z* 44 $(H_2N–CH_2–CH_2)^+$, *m*/*z* 58 $(H_2N–CH_2–CH_2–CH_2)^+$ and *m*/*z* 86 $(H_2N–CH_2–CH_2–CH_2–C=O)^+$, provide significant structural information. The $(H_2N–CH_2)^+$, $(H_2N–CH_2–CH_2)^+$ and $(H_2N– CH_2–CH_2–CH_2)^+$ fragment ions originate from the cleavage of the carbon-backbone, and $(H_2N–CH_2–CH_2–CH_2–C=O)^+$ fragment ions originate from the cleavage of C–O bond of the *γ*-aminobutyramide. The intensity distribution of their mapping images in Fig. 3b corresponds to low molecular mass compounds. Therefore, fragment ions at *m*/*z* 30, *m*/*z* 44, *m*/*z* 58 and *m*/*z* 86 can be assigned to *γ*-aminobutyramide. Moreover, since the peaks at *m*/*z* 56 and *m*/*z* 85 appeared in the mass

spectrum of γ-aminobutyramide by electron impact ionization [24], the secondary ion maps of *m/z* 56 and *m/z* 85 in Fig. 3b are also assigned to γ-aminobutyramide. In addition, *m/z* 30 $(H_2N–CH_2)^+$ represents fragment from the cleavage of the C–C bond of glycine, *m/z* 30 $(H_2N–CH_2)^+$ and *m/z* 44 $(H_2N–CH_2–CH_2)^+$ represent fragments from the cleavage of the carbon-backbone of β-alaninamide, *m/z* 30 $(H_2N–CH_2)^+$, *m/z* 44 $(H_2N–CH_2–CH_2)^+$ and *m/z* 58 $(H_2N–CH_2–CH_2–CH_2)^+$ represent fragments from the cleavage of the carbon-backbone of the *N*-acetylputrescine and putrescine, hence associated fragment ions can also be assigned to glycine, β-alaninamide, *N*-acetylputrescine and putrescine. On the other hand, the fragment ions at *m/z* 58 $((CH_3)_2N=CH_2)^+$, *m/z* 59 $((CH_3)_3N)^+$, and *m/z* 86 $((CH_3)_3N–CH=CH_2)^+$ represent the choline residues [25,26]. A peak found at a nominal mass of 104 *m/z* $((CH_3)_3N–CH_2–CH_2–OH)^+$ represents the molecular fragment of choline [25,26]. Since *m/z* 58 $((CH_3)_2N=CH_2)^+$ and *m/z* 59 $((CH_3)_3N)^+$ also represent the betaine residues, associated fragment ions can also be assigned to betaine. From the mapping images in Fig. 3b it is clear that the intensity distributions of *m/z* 58, *m/z* 59, *m/z* 86, and *m/z* 104 correspond to low molecular mass compounds. The fragment ion at *m/z* 84 $(C_4H_6NO)^+$ was assigned to the 2-pyrrolidone [27]. The $(C_4H_6NO)^+$ fragment ion originates from the cleavage of the N–H bond of the pyrrolidone ring.

In the secondary ion mass spectrum and mapped images of negative ions in Figs. 3d, 3e and 3f, the typical negative ions at *m/z* 16 O^-, *m/z* 17 $(OH)^-$ and *m/z* 32 O_2^- show a consistent distribution with low molecular mass compounds arising mainly from the inorganic and organic salt. The fragment ions at *m/z* 26 $(CN)^-$ and *m/z* 42 $(CNO)^-$ assigned to organic nitrogen. On the other hand, although fragment ions $(SO_3)^-$ and $(HPO_3)^-$ have an isobaric interference at *m/z* 80, the ion mapping at *m/z* 107 $(SO_3–CH_2–CH_2)^-$ supports that *m/z* 80 and *m/z* 107 are from *N*-acetyltaurine [28] and/or isethionic acid. The negative ion at *m/z* 33 $(SH)^-$ seems to come from the sulfur in *N*-acetyltaurine and/or isethionic acid and/or cysteine.

The fragment ions at *m/z* 31 P^-, *m/z* 47 $(PO)^-$, *m/z* 48 $(HPO)^-$, *m/z* 63 $(PO_2)^-$, *m/z* 64 $(HPO_2)^-$, *m/z* 79 $(PO_3)^-$, *m/z* 80 $(HPO_3)^-$ and *m/z* 81 $(H_2PO_3)^-$ assigned to dihydrogen phosphate. The dihydrogen phosphate